

Lab 11: Protein Structure Prediction with AlphaFold

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```
library(bio3d)
library(pheatmap)
library(jsonlite)
```

```
## Set Results Directory
results_dir <- "HIVPr_dimer_94b5b/"
```

```
## Read PDB Files
pdb_files <- list.files(
  path = results_dir,
  pattern = "*.pdb",
  full.names = TRUE
)

basename(pdb_files)
```

```
[1] "HIVPr_dimer_94b5b_unrelaxed_rank_001_alphafold2_ptm_model_4_seed_000.pdb"
```

```
[2] "HIVPr_dimer_94b5b_unrelaxed_rank_002_alphafold2_ptm_model_5_seed_000.pdb"
[3] "HIVPr_dimer_94b5b_unrelaxed_rank_003_alphafold2_ptm_model_3_seed_000.pdb"
[4] "HIVPr_dimer_94b5b_unrelaxed_rank_004_alphafold2_ptm_model_1_seed_000.pdb"
[5] "HIVPr_dimer_94b5b_unrelaxed_rank_005_alphafold2_ptm_model_2_seed_000.pdb"
```

```
## Align AlphaFold Models
```

```
pdbbs <- pdbaln(pdb_files, fit=TRUE, exefile="msa")
```

Reading PDB files:

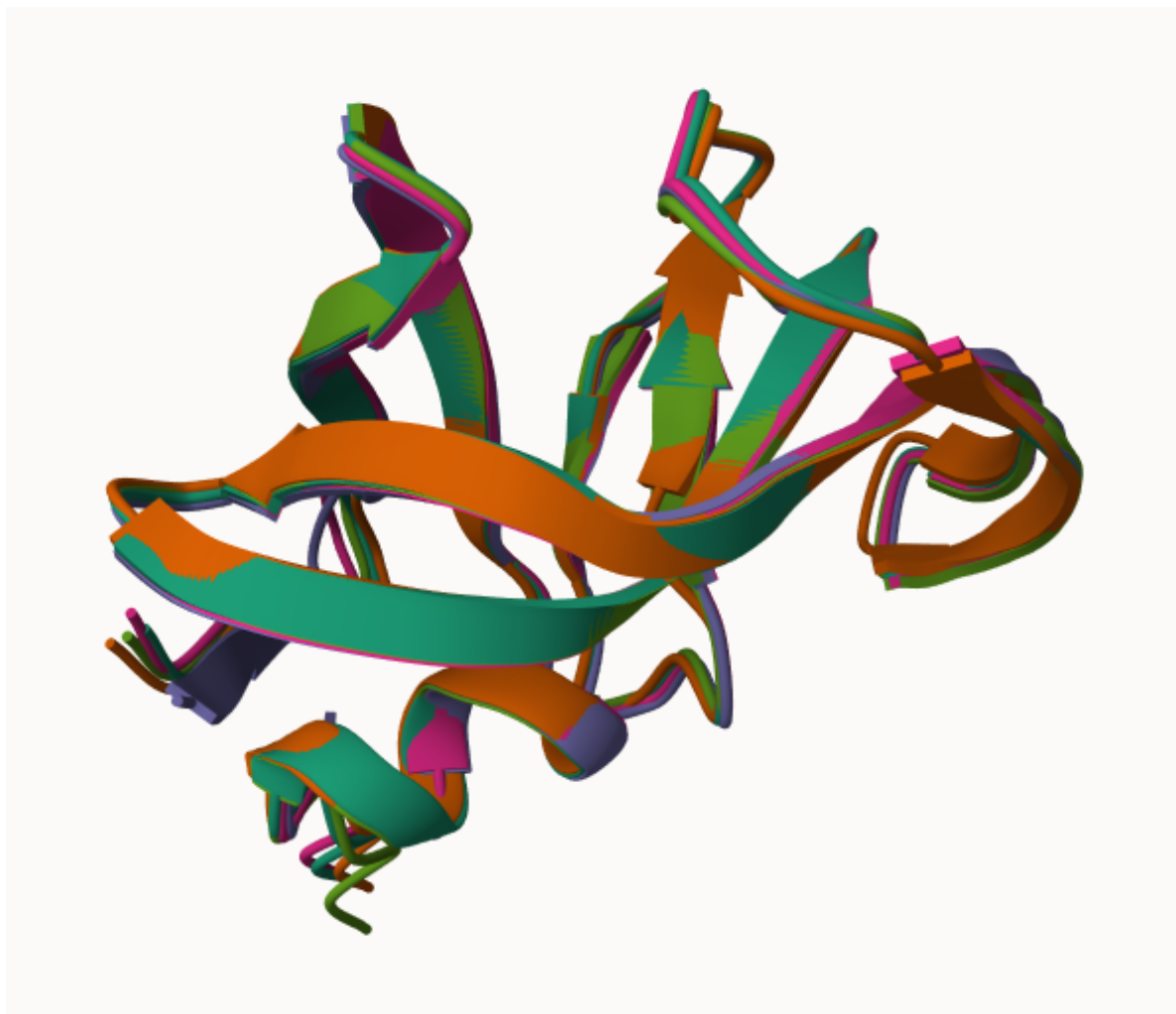
```
HIVPr_dimer_94b5b//HIVPr_dimer_94b5b_unrelaxed_rank_001_alphafold2_ptm_model_4_seed_000.pdb
HIVPr_dimer_94b5b//HIVPr_dimer_94b5b_unrelaxed_rank_002_alphafold2_ptm_model_5_seed_000.pdb
HIVPr_dimer_94b5b//HIVPr_dimer_94b5b_unrelaxed_rank_003_alphafold2_ptm_model_3_seed_000.pdb
HIVPr_dimer_94b5b//HIVPr_dimer_94b5b_unrelaxed_rank_004_alphafold2_ptm_model_1_seed_000.pdb
HIVPr_dimer_94b5b//HIVPr_dimer_94b5b_unrelaxed_rank_005_alphafold2_ptm_model_2_seed_000.pdb
.....
```

Extracting sequences

```
pdb/seq: 1   name: HIVPr_dimer_94b5b//HIVPr_dimer_94b5b_unrelaxed_rank_001_alphafold2_ptm_mo
pdb/seq: 2   name: HIVPr_dimer_94b5b//HIVPr_dimer_94b5b_unrelaxed_rank_002_alphafold2_ptm_mo
pdb/seq: 3   name: HIVPr_dimer_94b5b//HIVPr_dimer_94b5b_unrelaxed_rank_003_alphafold2_ptm_mo
pdb/seq: 4   name: HIVPr_dimer_94b5b//HIVPr_dimer_94b5b_unrelaxed_rank_004_alphafold2_ptm_mo
pdb/seq: 5   name: HIVPr_dimer_94b5b//HIVPr_dimer_94b5b_unrelaxed_rank_005_alphafold2_ptm_mo
```

The five AlphaFold models were aligned so their structures could be compared. This makes it easier to see how similar or different the predicted models are from each other.

Mol* Visualization of Superposed Models



The five AlphaFold models were superposed in Mol* to compare their structural similarity. Most regions overlap closely, indicating that the predicted models are generally consistent with each other.

Mol* Visualization Colored by pLDDT



The models were colored by pLDDT confidence score using the uncertainty/disorder setting in Mol*. Higher-confidence regions are predicted more reliably, while lower-confidence regions may be more flexible or uncertain.

Corefit Mol* Visualization



The core-fitted structures highlight the most structurally conserved regions between the models. Variable regions become easier to identify after fitting the stable core residues.

RMSD Analysis

```
rd <- rmsd(pdb, fit=TRUE)
```

Warning in rmsd(pdb, fit = TRUE): No indices provided, using the 99 non NA positions

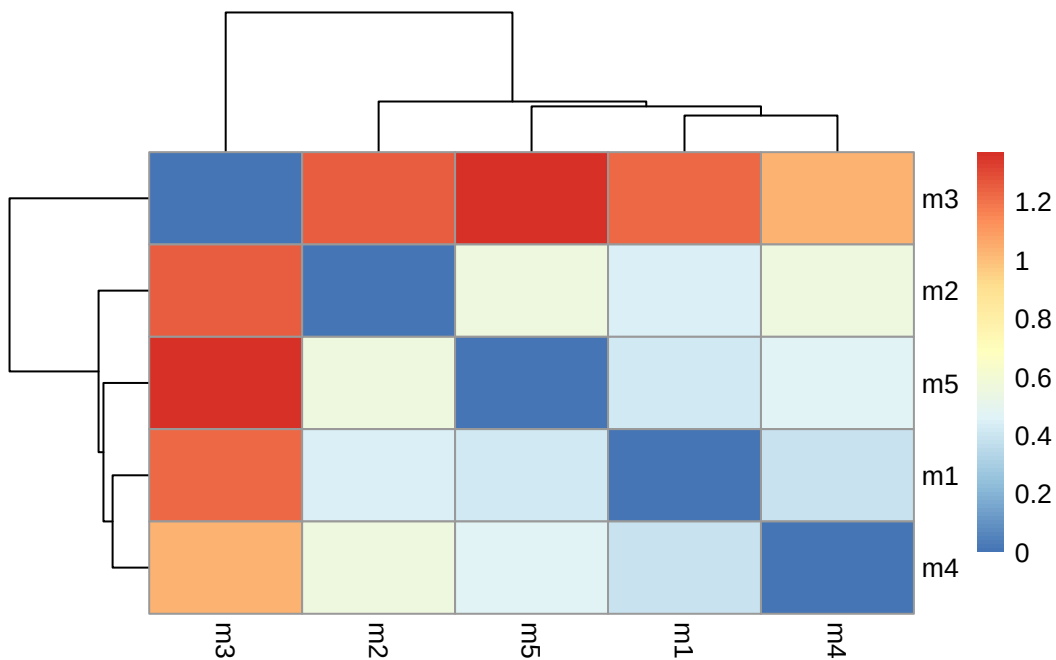
```
range(rd)
```

```
[1] 0.00 1.37
```

```
colnames(rd) <- paste0("m",1:5)
```

```
rownames(rd) <- paste0("m",1:5)
```

```
pheatmap(rd)
```



The RMSD heatmap compares structural similarity between the five AlphaFold models. Lower RMSD values indicate that two models are more structurally similar, while higher values indicate greater structural differences.

pLDDT Confidence Analysis

```
pdb <- read.pdb("1hsg")
```

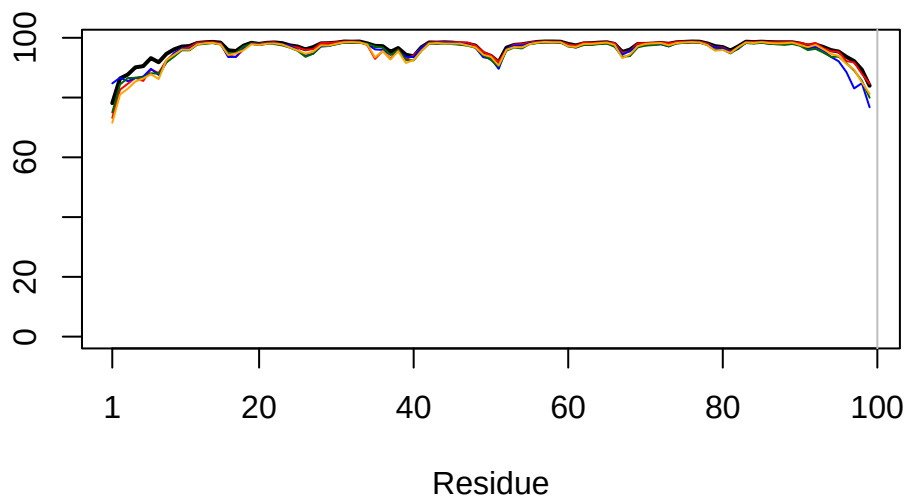
Note: Accessing on-line PDB file

```
plotb3(pdb$b[1,], typ="l", lwd=2, sse=pdb)
```

Warning in plotb3(pdb\$b[1,], typ = "l", lwd = 2, sse = pdb): Length of input 'sse' does not equal the length of input 'x'; Ignoring 'sse'

```
points(pdb$b[2,], typ="l", col="red")
points(pdb$b[3,], typ="l", col="blue")
points(pdb$b[4,], typ="l", col="darkgreen")
points(pdb$b[5,], typ="l", col="orange")

abline(v=100, col="gray")
```



The pLDDT plot shows AlphaFold confidence scores across residue positions for each model. Higher scores indicate more reliable predicted structures, while lower scores indicate less confident regions.

Core Fitting

```
core <- core.find(pdb)
```

```
core size 98 of 99  vol = 4.376
core size 97 of 99  vol = 3.604
core size 96 of 99  vol = 2.86
core size 95 of 99  vol = 2.352
core size 94 of 99  vol = 1.794
core size 93 of 99  vol = 1.303
core size 92 of 99  vol = 0.946
core size 91 of 99  vol = 0.579
core size 90 of 99  vol = 0.351
FINISHED: Min vol ( 0.5 ) reached
```

```
core.inds <- print(core, vol=0.5)
```

```
# 91 positions (cumulative volume <= 0.5 Angstrom^3)
  start end length
1      3   3      1
2      7  96     90
```

```
xyz <- pdbfit(
  pdbs,
  core.inds,
  outpath="corefit_structures"
)
```

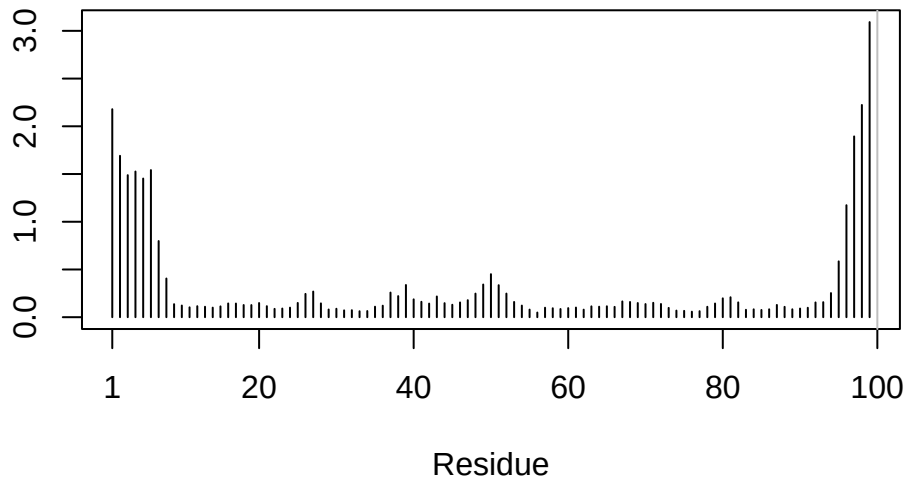
Core fitting identifies the most structurally conserved regions shared across the models. These conserved regions are then used to improve structural alignment between predictions.

RMSF Analysis

```
rf <- rmsf(xyz)
plotb3(rf, sse=pdb)
```


Warning in plotb3(rf, sse = pdb): Length of input 'sse' does not equal the length of input 'x'; Ignoring 'sse'

```
abline(v=100, col="gray")
```

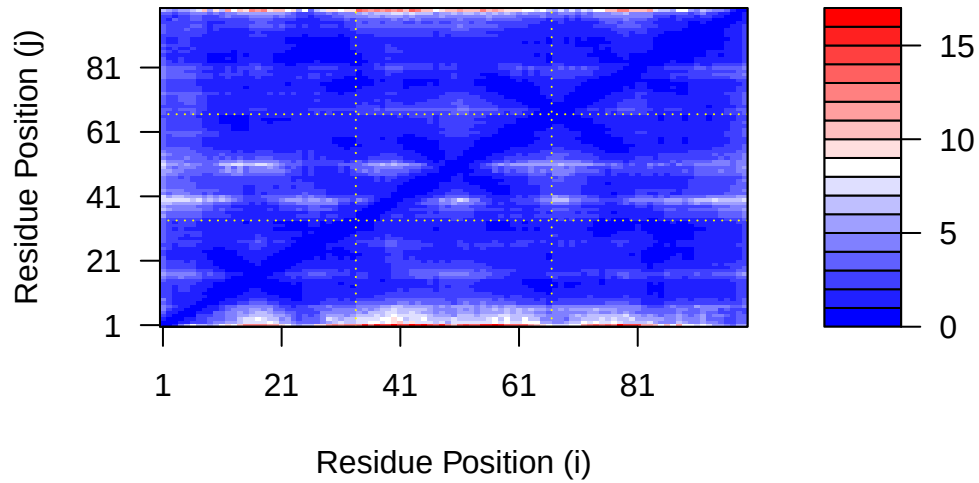


The RMSF plot shows how much each residue position varies across the five AlphaFold models. Higher RMSF values indicate more flexible or variable regions of the protein.

PAE Analysis

```
pae_files <- list.files(  
  path=results_dir,  
  pattern=".*model.*\\.json",  
  full.names=TRUE  
)  
  
pae1 <- read_json(pae_files[1], simplifyVector=TRUE)  
  
plot.dmat(  
  pae1$pae,  
  xlab="Residue Position (i)",
```

```
ylab="Residue Position (j)"  
)
```



The PAE plot shows AlphaFold confidence in the relative positioning of residues within the structure. Lower PAE values indicate more reliable positioning, while higher values indicate greater uncertainty.